Band-1 Graph Isomorphism: Last Log-Factors Probe (Collapse List 2.15)

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Honesty stamp. Babai's algorithm places general Graph Isomorphism (GI) in **quasipolynomial** time after the 2017 Split-or-Johnson fix confirmed by Helfgott. This note does **not** claim GI $\in$ P. **GATE-04** is locked out of scope. **No Clay** stamp (GI is not a Clay Millennium problem). No ToE stamp.

> GI $\in$ P **remains OPEN.** Best accepted *general unconditional* upper bound is still Babai quasipolynomial $2^{O((\log n)^c)}$ (Helfgott exposition takes $c = 3$). The live incremental program is to remove the remaining log-power gap down to polynomial.

Abstract

Literature ledger (1982–2026 sweep) cites Luks, Babai–Luks / Zemlyachenko, Babai 2015, the January 2017 Helfgott error + Babai fix, Helfgott Bourbaki / Astérisque 407, and Babai STOC 2019 canonization (preliminary). **Cited best accepted general form:** quasipolynomial $2^{O((\log n)^c)}$ with Helfgott's $c = 3$ reading. No peer-reviewed unconditional general **polynomial-time** algorithm for all graphs was elevated in the 2024–2026 sweep; heuristic / conditional P claims are ledger-noted only. Box work: tiny WL-1 + backtracking isomorphism on ER graphs $n \leq 12$ — **method demo only**, not Babai and not a complexity claim.

Problem (collapse 2.15)

Graph Isomorphism: given undirected graphs G, H on n vertices, decide whether there exists a bijection of vertices preserving adjacency.

- Classical complexity: in NP $\cap$ co-AM; not known to be NP-complete (unless PH collapses); defines the class GI.
- **Band target:** place general GI in P (remove the last log-factors after quasipolynomial).
- **Honest present tense:** quasipolynomial is accepted; polynomial is **OPEN**.

Literature ledger (2024–2026 status; cite carefully)

Year	Who	Bound / event (schematic)	Status
1982	Luks	P for bounded degree; SI group framework	classical foundation
1983	Babai–Luks / Zemlyachenko	$2^{O(\sqrt{n \log n})}$ general	~30-year SOTA

Year	Who	Bound / event (schematic)	Status
2015	**Babai**	quasipolynomial $\exp((\log n)^{O(1)})$ announced	arXiv:1512.03547
2017-01-04	Helfgott / Babai	Split-or-Johnson analysis error → temporary retract of quasipoly claim (still subexp)	Quanta; Babai update
2017-01-09/19	**Babai fix**	quasipolynomial **restored**	Babai update; Quanta
2017 / 2019	**Helfgott**	Bourbaki exposition; claims $c = 3 \Rightarrow 2^{O((\log n)^3)}$	arXiv:1701.04372; 1710.04574; *Astérisque* 407 Exp. 1125
2019	Babai	quasipolynomial **canonization** preliminary report	STOC 2019
2024–2026	community / heuristics	no elevated unconditional general P algorithm	Wikipedia SOTA note (as of 2025): full Babai writeups still incomplete as final monographs in places — **does not reopen** quasipoly; DRESS / heat-kernel heuristics **not** elevated to P

Canonical citations.

1. L. Babai, *Graph Isomorphism in Quasipolynomial Time*, arXiv:1512.03547.
2. H. A. Helfgott, *Isomorphismes de graphes en temps quasi-polynomial (d'après Babai)*, Séminaire Bourbaki; arXiv:1701.04372 / 1710.04574; *Astérisque* **407** (2019), Exp. 1125, pp. 135–182.
3. L. Babai, *Canonical form for graphs in quasipolynomial time: preliminary report*, STOC 2019.

Hygiene — Babai fix / Helfgott / subsequent (accurate).

- The January 2017 episode was a **real error** in the Split-or-Johnson recursive analysis, briefly demoting the *claimed* bound to subexponential; Babai's **replacement recursion** restored quasipolynomial within days; Helfgott **publicly confirmed** the fix (Bourbaki).
- Community acceptance of quasipolynomial rests primarily on Helfgott's exposition + absence of further known fatal gaps (cstheory consensus).
- As of 2025–26, some notes still flag that Babai's full polished monograph / complete published writeup of all pieces is unfinished; that archival gap is **not** a license to claim the result failed, nor to claim GI ∈ P.
- Subsequent 2024–2026 arXiv heuristics (e.g. dynamical / diffusion refinements) are **empirical or conditional** and are **not** elevated here as placing GI in P.

Method (box) — illustration only

1. Build ER graphs $G \sim G(n, 0.35)$ for $n \in \{6, 8, 10, 12\}$.
2. Positive: random permutation of vertices ⇒ isomorphic twin.
3. Negative: delete one edge (when present) ⇒ typically non-isomorphic.
4. Decide isomorphism by **1-dimensional color refinement** + **backtracking** with color-class domains (pure Python; $n \leq 12$ only).

Label: textbook tiny solver — **not** Babai local certificates / Johnson obstruction machinery, and **not** a timing claim about quasipolynomial vs polynomial.

Results

Unit tests

PASS: path$_3 \not\simeq K_3$; $K_3 \simeq K_3$; $C_4 \not\simeq 2K_2$; $C_4 \simeq$ permuted C_4.

Random trials (method demo)

n	trials	isomorphic twins OK	edge-flip non-iso OK	typical t (s)
6	5	5/5	5/5	$< 10^{-4}$
8	5	5/5	5/5	$< 10^{-4}$
10	5	5/5	5/5	$< 10^{-4}$
12	5	5/5	5/5	$< 10^{-4}$

Aggregate: 20/20 checks OK; wall-clock full demo ≈ 0.002 s. **Label:** finite method sanity — **zero** complexity consequence for the log-factor program.

Reproducibility

- Lab: `/workspace/mission-control/experiments/band1/graph_iso_logfactors/`
- Run: `python3 iso_small_demo.py`
- Artifacts: `results.json`, `literature_ledger.json`, `SUMMARY.txt`, `METHOD.md`
- Paper: `papers/2026-09-11_Band1_graph_iso_logfactors_probe.md`

Conclusion

Item	Status
Babai quasipolynomial (post-2017 fix)	**ACCEPTED** community status; Helfgott $c = 3$ reading
Helfgott Bourbaki / Astérisque 407	elevated exposition / validation
GI $\in$ P / last log-factors	**OPEN** (collapse 2.15 live)
Box iso demo	illustration only
GATE-04	**LOCKED**
Clay / ToE	**not claimed**

Next open named (incremental). Any peer-reviewed improvement that (i) lowers the quasipolynomial exponent c, or (ii) places a nontrivial broader class into P on the way to full GI $\in$ P, or (iii) a complete final monograph of Babai's algorithm — without pretending polynomial time is already proved.